Existence and Uniqueness for ODEs: The Picard-Lindelöf Theorem

Math Foundations — Node 30

First, an example

Before the general theorem, look at the simplest nontrivial IVP: $y' = y$ with $y(0) = 1$. Here $f(t, y) = y$, so $|f(t, y_1) - f(t, y_2)| = |y_1 - y_2|$ and f is Lipschitz in y with constant $L = 1$. The Picard iterates starting from $y_0(t) \equiv 1$ are

$$y_1(t) = 1 + \int_0^t 1\, ds = 1 + t, \qquad y_2(t) = 1 + \int_0^t (1 + s)\, ds = 1 + t + \tfrac{t^2}{2},$$

Read aloud: "y-one of t equals one plus the integral from zero to t of one d-s, which is one plus t; y-two of t equals one plus the integral from zero to t of one plus s d-s, which is one plus t plus t-squared over two." and inductively $y_k(t) = \sum_{j=0}^{k} t^j/j!$, the partial sums of the Taylor series of e^t. They converge uniformly on every compact interval to the unique solution $y(t) = e^t$.

Contrast this with $y' = y^{2/3}$, $y(0) = 0$. Now $\partial_y f = \frac{2}{3} y^{-1/3}$ blows up at 0, so f is *not* Lipschitz near $y = 0$, and the IVP admits both $y \equiv 0$ and $y(t) = (t/3)^3$ as solutions. Lipschitz continuity is exactly what rules out this branching.

1 Setting

Let $D \subset \mathbb{R} \times \mathbb{R}^n$ be an open set, $(t_0, y_0) \in D$, and $f : D \to \mathbb{R}^n$ a continuous function. We study the initial value problem (IVP)

$$y'(t) = f(t, y(t)), \qquad y(t_0) = y_0. \tag{1}$$

Read aloud: "y-prime of t equals f of t and y of t, with y of t-naught equal to y-naught." A *solution* is a differentiable curve $y : I \to \mathbb{R}^n$ on an open interval $I \ni t_0$ satisfying (1) pointwise.

2 Lipschitz continuity and Picard iteration

Definition 1 (Lipschitz in y)**.** A function $f : D \to \mathbb{R}^n$ is *Lipschitz continuous in y* on D if there is $L \geq 0$ such that

$$\|f(t, y_1) - f(t, y_2)\| \leq L\, \|y_1 - y_2\| \qquad \forall (t, y_1), (t, y_2) \in D.$$

Read aloud: "the norm of f of t y-one minus f of t y-two is at most L times the norm of y-one minus y-two, for all pairs in D." The smallest such L is the *Lipschitz constant* of f.

Remark 2. If f is C^1 on D, then $\partial f/\partial y$ being bounded by L on D (in operator norm) implies f is Lipschitz in y with constant L. This is the most common way Lipschitz hypotheses arise in practice.

Definition 3 (Picard iteration)**.** Given the IVP (1), define the sequence of functions $\{y_k\}_{k\geq 0}$ by

$$y_0(t) \equiv y_0, \qquad y_{k+1}(t) \;=\; y_0 + \int_{t_0}^{t} f\big(s, y_k(s)\big)\, ds.$$

Read aloud: "y-naught of t is identically y-naught; y-sub-k-plus-one of t equals y-naught plus the integral from t-naught to t of f of s and y-k of s, d-s." These are the *Picard iterates* of the IVP.

A continuous function y solves (1) if and only if it is a fixed point of the integral operator $T[y](t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds$. So existence and uniqueness become a fixed-point problem for T.

3 The Picard-Lindelöf theorem

Theorem 4 (Picard-Lindelöf)**.** *Let $a, b > 0$ and let*

$$R = \{(t, y) \in \mathbb{R} \times \mathbb{R}^n : |t - t_0| \leq a,\ \|y - y_0\| \leq b\}.$$

Read aloud: *"R is the set of pairs t comma y in R-cross-R-n such that the absolute value of t minus t-naught is at most a, and the norm of y minus y-naught is at most b." Suppose f is continuous on R with $\|f(t, y)\| \leq M$ for all $(t, y) \in R$, and Lipschitz in y on R with constant L. Set*

$$\alpha = \min\big(a,\ \tfrac{b}{M}\big)\,.$$

Read aloud: *"alpha is the minimum of a and b-over-M." Then the IVP* (1) *has a unique solution $y : [t_0 - \alpha,\ t_0 + \alpha] \to \mathbb{R}^n$, and the Picard iterates y_k converge to it uniformly on that interval.*

Proof sketch. Set $I = [t_0 - \alpha, t_0 + \alpha]$ and let

$$X = \{y \in C(I, \mathbb{R}^n) : \|y(t) - y_0\| \leq b \text{ for all } t \in I\}.$$

Read aloud: "X is the set of continuous functions y from I into R-n such that the norm of y of t minus y-naught is at most b for all t in I." Equip $C(I, \mathbb{R}^n)$ with the sup norm $\|y\|_\infty = \sup_{t\in I} \|y(t)\|$. Then X is a closed subset of a Banach space, hence complete. Define $T : X \to C(I, \mathbb{R}^n)$ by

$$T[y](t) = y_0 + \int_{t_0}^{t} f\big(s, y(s)\big)\, ds.$$

Read aloud: "T applied to y, evaluated at t, equals y-naught plus the integral from t-naught to t of f of s and y of s, d-s."

Step 1 (T maps X into X). For $y \in X$ and $t \in I$,

$$\|T[y](t) - y_0\| \leq \int_{t_0}^{t} \|f(s, y(s))\|\, ds \leq M|t - t_0| \leq M\alpha \leq b.$$

Read aloud: "the norm of T-of-y at t minus y-naught is at most the integral of the norm of f, which is at most M times t minus t-naught, which is at most M-alpha, which is at most b." Hence $T[y] \in X$.

Step 2 (Contraction in a weighted norm). On $C(I, \mathbb{R}^n)$ define the equivalent norm $\|y\|_w = \sup_{t\in I} e^{-2L|t-t_0|}\, \|y(t)\|$. Then for $y_1, y_2 \in X$ and $t \geq t_0$ in I,

$$\begin{aligned}
\|T[y_1](t) - T[y_2](t)\| &\leq \int_{t_0}^{t} \|f(s, y_1(s)) - f(s, y_2(s))\|\, ds \\
&\leq L \int_{t_0}^{t} \|y_1(s) - y_2(s)\|\, ds \\
&\leq L \int_{t_0}^{t} e^{2L(s-t_0)}\, ds \cdot \|y_1 - y_2\|_w \\
&= \tfrac{1}{2}\left(e^{2L(t-t_0)} - 1\right) \|y_1 - y_2\|_w.
\end{aligned}$$

Read aloud: "the norm of T-of-y-one minus T-of-y-two at t is bounded by the integral of the difference of f's, then by L times the integral of the difference of y's, then by L times the integral of e-to-the-two-L-s-minus-t-naught d-s times the weighted norm, equaling one-half times e-to-the-two-L-t-minus-t-naught minus one, times the weighted norm." Multiplying by $e^{-2L(t-t_0)}$ and taking sup over t (the $t < t_0$ case is symmetric) yields

$$\|T[y_1] - T[y_2]\|_w \leq \tfrac{1}{2}\,\|y_1 - y_2\|_w.$$

Read aloud: "the weighted norm of T-of-y-one minus T-of-y-two is at most one-half times the weighted norm of y-one minus y-two." Thus T is a strict contraction on $(X, \|\cdot\|_w)$ with contraction factor 1/2.

Step 3 (Banach fixed point). By the Banach contraction mapping theorem, T has a unique fixed point $y^* \in X$. By construction y^* is continuous and satisfies the integral equation; since f is continuous, y^* is in fact C^1 and solves (1) pointwise on I. Uniqueness in X follows from the contraction property; any solution must lie in X (by Step 1 applied to it), so uniqueness is global on I.

Finally, the Picard iterates are exactly the sequence $T^k[y_0]$ produced by the contraction mapping theorem, and so converge uniformly on I to y^*. □

4 What goes wrong without Lipschitz

Example 5 (Non-uniqueness)**.** Consider $y' = y^{2/3}$, $y(0) = 0$. Here $f(t, y) = y^{2/3}$ fails to be Lipschitz at $y = 0$ since $\partial_y f = \frac{2}{3}y^{-1/3}$ blows up. The IVP admits the two distinct solutions $y \equiv 0$ and $y(t) = (t/3)^3$, as well as a one-parameter family y_τ that equals 0 for $t \leq \tau$ and $((t-\tau)/3)^3$ for $t > \tau$.

5 Connection to flow-based generative models

Modern generative models (flow matching, ELF, neural ODEs) define

$$\dot{x}(t) = v_\theta(t, x(t))$$

Read aloud: "x-dot of t equals v-sub-theta of t and x of t." where v_θ is a neural network. On any compact set K, neural networks built from smooth activations are C^1 and hence Lipschitz in x with a finite constant L_K. Theorem 4 therefore guarantees a unique sample trajectory for each starting point x_0, and hence a well-defined flow ϕ_t on K. Smooth dependence on initial conditions further makes ϕ_t a diffeomorphism — a property used implicitly throughout the literature.